Ingineria programării

Curs 11 – 27 Mai

Outline

- Recap – Software testing
- Non-functional software testing
  - Security
  - Performance
  - Usability
  - Internationalization
- Refactoring
- Software Estimation

Correctness

- Correctness of an algorithm is asserted when it is said that **the algorithm is correct with respect to a specification**
- **Functional correctness** refers to the input–output behavior of the algorithm (i.e., for each input it produces the correct output)
- A distinction is made between **total correctness**, which additionally requires that the algorithm terminates, and **partial correctness**, which simply requires that *if* an answer is returned it will be correct.

Non Functional Software Testing

- Verifies that the software functions properly even when it receives invalid or unexpected inputs
- Methods:
  - **Performance testing** or **Load Testing** checks to see if the software can handle large quantities of data or users (software scalability).
  - **Usability testing** checks if the user interface is easy to use and understand.
  - **Security testing** is essential for software which processes confidential data and to prevent system intrusion by hackers.
  - **Internationalization and localization** is needed to test these aspects of software, for which a pseudo localization method can be used.

Software Performance Testing

- Types
  - **load testing** – can be the expected concurrent number of users on the application (database is monitored)
  - **stress testing** – is used to break the application (2 x users, extreme load) (application's robustness)
  - **endurance testing** – if the application can sustain the continuous expected load (for memory leaks)
  - **spike testing** – spiking the number of users and understanding the behavior of the application whether it will go down or will it be able to handle dramatic changes in load

Performance Testing Analysis

Usability testing

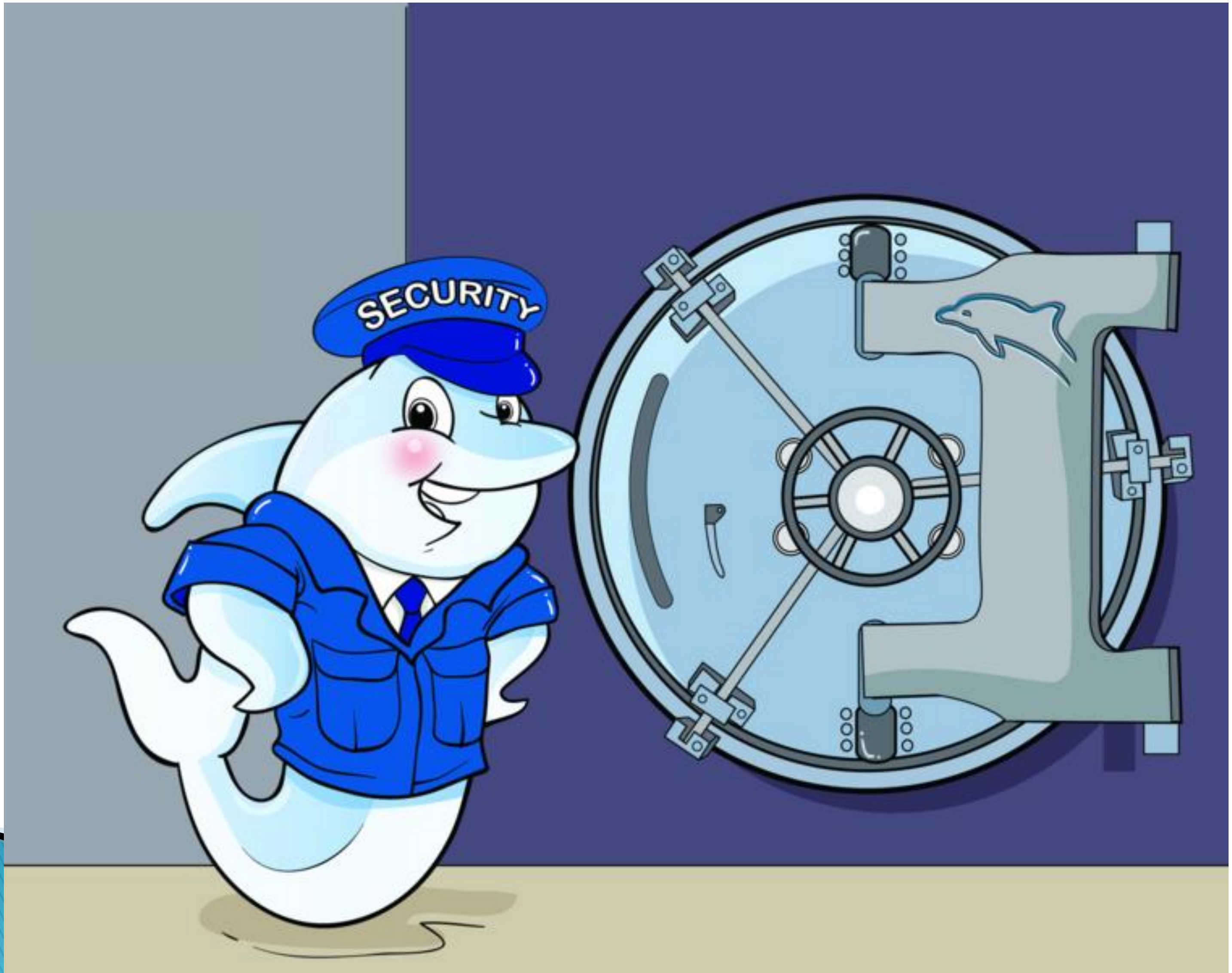
- A technique used to evaluate a product by testing it on users
- Usability testing focuses on measuring a human-made product's capacity to meet its intended purpose.
- **Examples** of products that commonly benefit from usability testing are web sites or web applications, computer interfaces, documents, or devices
- **Goals**
  - *Performance* – How much time, steps?
  - *Accuracy* – How many mistakes did people make?
  - *Recall* – How much does the person remember afterwards or after periods of non-use?
  - *Emotional response* – How does the person feel about the tasks completed? Is the person confident, stressed? Would the user recommend this system to a friend?

Usability testing steps

Security testing

- The Process to determine that an Information System protects data and maintains functionality as intended.
- The six basic security concepts that need to be covered by security testing are:
  - **Confidentiality**,
  - **Integrity** – information which it receives has not been altered in transit or by other than the originator of the information
  - **Authentication** – validity of a transmission, message, or originator,
  - **Authorization** – determining that a requester is allowed to receive a service or perform an operation,
  - **Availability** – Assuring information and communications services will be ready for use when expected,
  - **Non-repudiation** – prevent the later denial that an action happened, or a communication that took place

Security logo



Internationalization and localization

- Means of adapting computer software to different languages and regional differences
- **Internationalization** is the process of designing a software application so that it can be adapted to various languages and regions without engineering changes.
- **Localization** is the process of adapting software for a specific region or language by adding locale-specific components and translating text.

Measuring software testing

- Usually, quality is constrained to such topics as **correctness, completeness, security**
- Can also include capability, reliability, efficiency, portability, maintainability, compatibility, and usability
- There are a number of common software measures, often called "metrics", which are used to measure the state of the software or the adequacy of the testing.

Degradation of code

- Initial code is clean and well structured
- As it is developed, it degrades because of incremental changes
- Eventually, it becomes difficult to maintain, extend or debug
  - Rigid, Fragile, Immobile, Viscous
    - Design Principles and Design Patterns. Robert C. Martin

Degradation of code

- **Rigidity** – the tendency for software to be difficult to change, even in simple ways
- **Fragility** – the tendency of the software to break in many places every time it is changed
- **Immobility** – the inability to reuse software from other projects or from parts of the same project
- **Viscosity** – it is easy to do the wrong thing, but hard to do the right thing

Refactoring

- "… *the process of changing a software system in such a way that it does not alter the external behavior of the code yet improves its internal structure*." – Martin Fowler in *Refactoring Improving The Design Of Existing Code*
- Making code cleaner, simpler and more elegant
- The refactored program must be functionally equivalent to the initial program

Refactoring

- Refactoring doesn’t prevent changing functionality, it just says that it’s a different activity from rearranging code
- It is easier to improve code quality if you don’t also change its functionality
- Refactoring is not rewriting
  - refactoring does not change functionality
  - rewriting does change it

Why to Refactor?

- Increase cohesion and decrease coupling
- Refactored code is easier to maintain and extend
- Simpler to use design patterns
- Improves performance and usability of code

When is Refactoring Necessary?

- Duplicate Code
- Long Methods
- Large classes
- Long lists of parameters
- Instructions switch
- Speculative generality
- Intense communication between objects
- Chaining Messages

How to Refactor

- Prepare a set of automatic tests for the program to be refactored
- Change the code in small iterations
  - Test for each iteration to ensure semantic equivalence
  - If any test fails, undo the change and try a different one
- After numerous iterations the cumulated changes will be large

Refactoring Techniques

- Increase abstraction
  - **Encapsulate Field** – force code to access the field with getter and setter methods
  - **Generalize Type** – create more general types to allow for more code sharing
  - **Replace type** – checking code with State/Strategy
  - **Replace conditional** with polymorphism
- Split code into more logical pieces
  - **Extract Method**, to turn part of a larger method into a new method.
  - **Extract Class** moves part of the code from an existing class into a new class.

Refactoring Techniques

- **Improving names and location of code**
  - **Move Method** or **Move Field** – move to a more appropriate Class or source file
  - **Rename Method** or **Rename Field** – changing the name into a more representative new one
  - **Pull Up** – in OOP, move to a superclass
  - **Push Down** – in OOP, move to a subclass

Software Quality (1)

- We do not assess construction quality (=> unique among engineering applications)
- All quality attributes refer to design.
- Esthetic qualities:
  - Software is mostly invisible, and esthetics only matter for the visible elements
  - Apart from the GUI, observable aspects software are:
    - Notations for design and writing of code
    - Behavior of software when interacting with other entities.

Software Quality (2)

- When discussing software quality we must:
  - **Define those attributes of quality** that are of interest;
  - Determine a way of **measuring those attributes**;
  - Find a way of **representing design**;
  - **Write specifications** that will guide developers (following and implementing design qualities).

Source Code

- Code that implements a given design is a representation of that design.
- Performing quality assurance after writing code is expensive and possibly useless.
- Usually, only the manner in which the code is written is taken into account (coding style, design patterns, adaptability, maintenance, reuse (coupling, cohesion), security)

Software Quality (3)

- Measures the appropriateness of software to the environment it is used in.

- Variuos aspecte taken into account are:
  - The software is running;
  - The software performs according to specifications;
  - The software is safe;
  - The software can be adapted as requirements change.

- All measurements regarding quality are relative!

Aspects of Software Quality

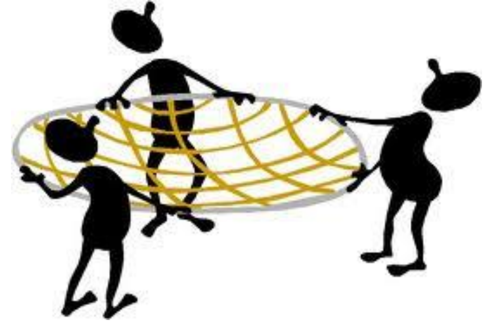
- Safety

- Efficiency

- Maintenance

- Usability

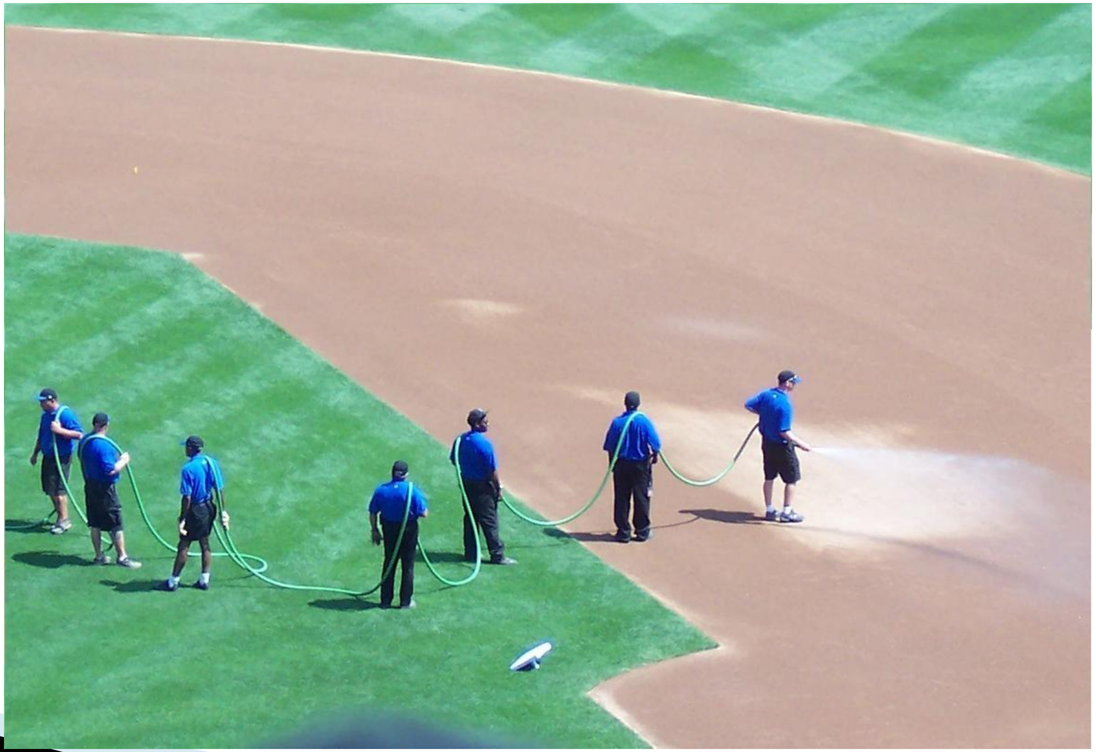
Safety



- Is the software **complete, correct and robust**?
  - **Completeness** – works for all possible inputs;
  - **Consistency** – always behaves as expected;
  - **Robustness** – behaves well in abnormal situations (eg. Lack of resources, lack of internet connection, etc.)

Efficiency

- The software makes efficient use of available resources (CPU time, network connection, etc.)
- Efficiency is always less important than safety. It is easier to make safe software efficient than the reverse



Maintenance

- How easily can the design be changed or adapted?

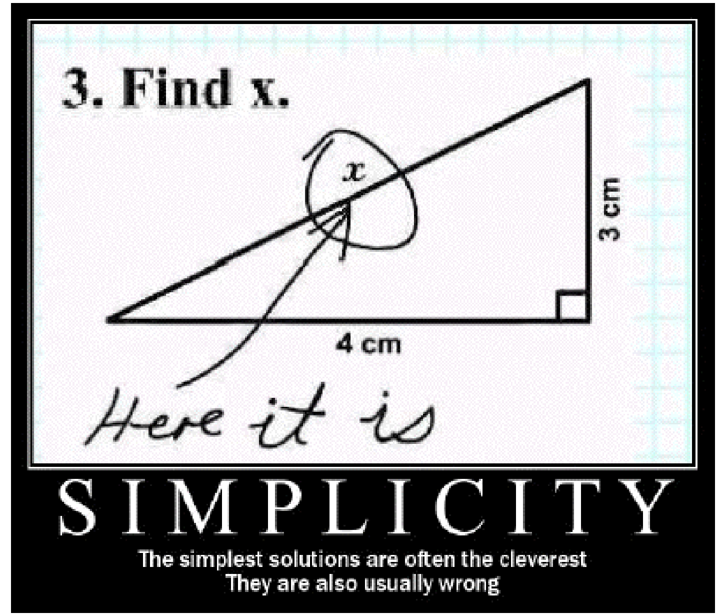
- Types of maintenance:
  - **Corrective**: error fixing;
  - **Perfective**: adding features that should have been part of the product;
  - **Adaptive**: updating software as requirements change.

Usability

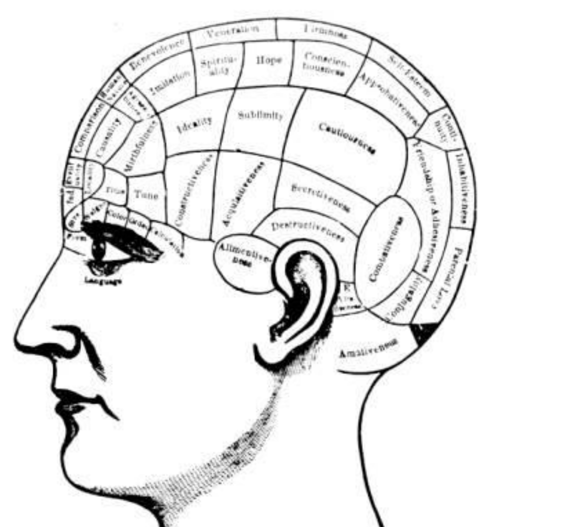
- How easily can the software be taught and used?

Measurable Attributes

- Simplicity



- Modularity

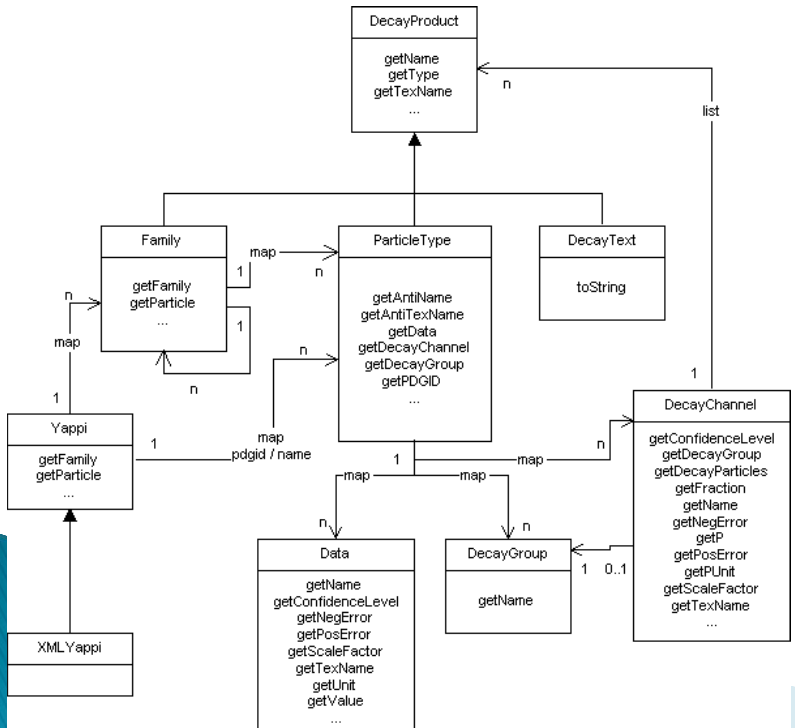


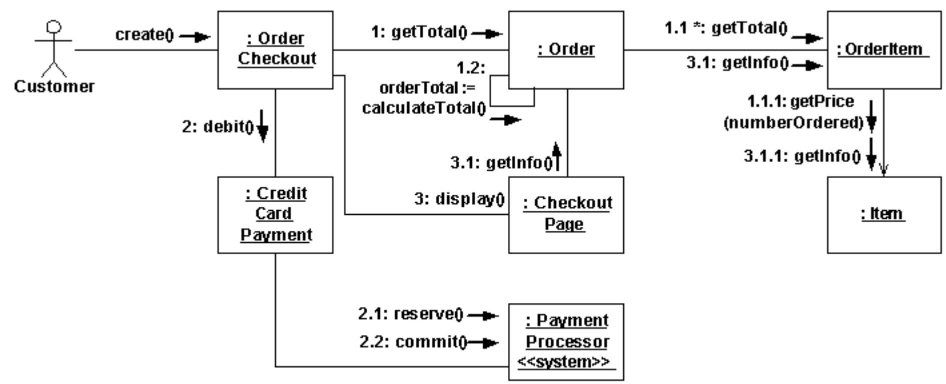
Simplicity

- The reverse of complexity.
- Aspects of complexity:
  - **Control flow**: counts all the possible execution paths for a program
  - **Information flow**: measures amount of data transmitted within the program
  - **Understanding**: counts the number of identifiers and operators

Modularity

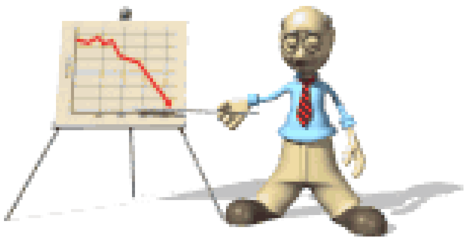
- Can be measured by examining :
  - **Cohesion**: how well the components of a module collaborate.
  - **Coupling**: interaction between modules





Why Use Metrics?

- We use metrics to
  - understand
  - control
  - predict



What Should We Measure

- Size of software
- Complexity of software
- Robustness of software
- Amount of time required to develop some software
- Resource allocation for development
- Productivity of effort
- Development costs

Estimation

- Intuitively, estimation seem subjective
  - To inexperienced persons, it looks like predicting the future
  - This is reinforced when estimation is incorrect and projects are delivered late
- Formal estimation processes
  - allows the project team to reach a consensus on the estimates
  - improve the accuracy

Estimation

- Successful estimations take into account the following
  - Work Breakdown Structure (WBS) – what are the tasks that need to be performed to finish the product?
  - Assumptions – how to deal with incomplete information
  - Trust – if stakeholders and engineers trust each other, the estimate will be more accurate

Work Breakdown Structure

- A list of tasks that, if completed, will produce the final product
  - Broken down by feature
  - By project phase (requirements tasks, design tasks, programming tasks, QA tasks, etc.)
  - Some combination of the two
  - Should reflect the way previous projects have been developed

Work Breakdown Structure

- A project should be broken down into 10 – 20 tasks
  - Regardless of the size of the project
  - For large projects (e.g. an operating system), the tasks are large
  - For smaller projects, the tasks are correspondingly smaller
- Create an estimate for the cost of each task
  - Most accurate estimates are those that rely on prior experience
  - NO estimate is guaranteed to be accurate

Assumptions

- At the beginning of the development team members do not have all the information
  - Assumptions are needed to fill in missing things
  - Assumptions can also be placeholders which will be corrected later
  - If an assumption is proven incorrect, the timeline of the project MUST be adjusted
- For effective estimates, assumptions need to be written down
  - If not, the team will need to have the same discussion again

Assumptions and Trust

- Estimates can either be a source of trust or distrust between the project team and managers.
- Stakeholders need the project completed but usually do not have software engineering experience
- Project managers must take care to make the estimation process as open and honest as possible

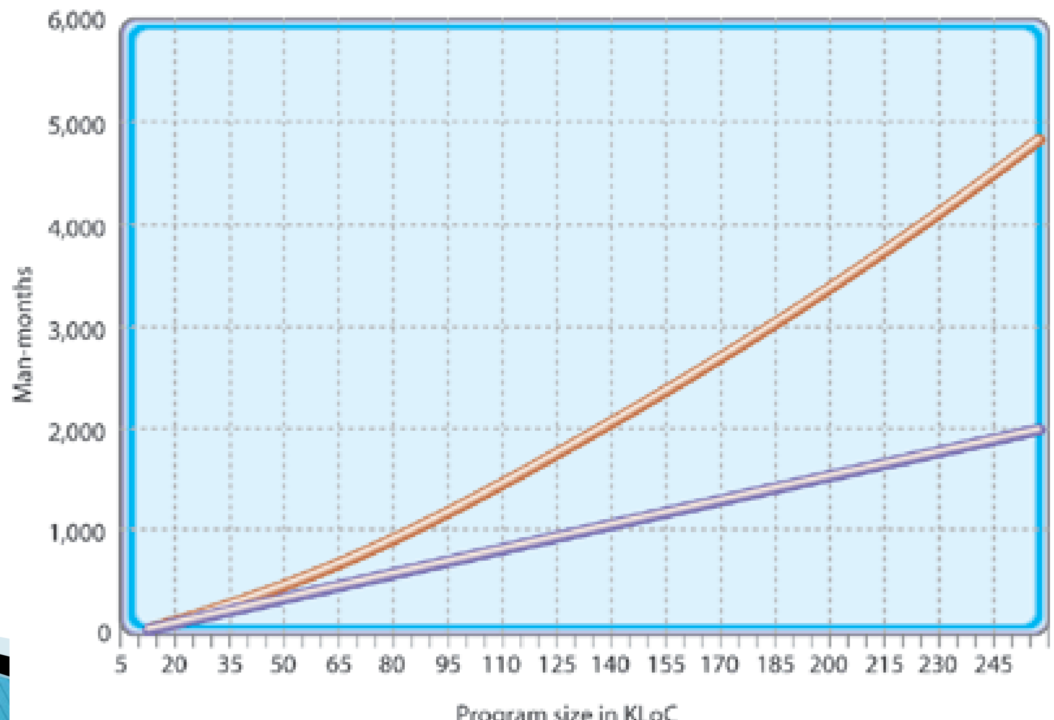
Assumptions and Trust

- It is common for nontechnical people to assume that programmers pad their estimates
  - They have a “rule” by which they cut off a third or half of any estimate
  - This lack of trust causes engineers to automatically pad their estimates
- An important part of running successful software projects is reaching a common understanding between the engineers, managers, and stakeholders.

Basic Metrics

- KLOC: Kilo Lines Of Code

- Effort, PM: Person – Month



COCOMO 2

- Boehm 1995
- Takes into consideration high level development tools and techniques
  - Prototyping
  - Modular development
  - 4GL (fourth generation language)
- Allows for estimates from the very first stages of development

COCOMO 2: Initial prototyping

- Effort required to create a prototype of the application
- Based on the Number of Object Points (NOP)
- Formula for computing effort:

$$PM = (1 - P_{reuse})\frac{NOP}{PROD}$$

NOP

- Investigate the screens and dialogs that are needed
  - Simple: 1
  - Complex: 2
  - Very complex: 3
- Reports that need generated
  - Simple: 2
  - Complex: 5
  - Very complex: 8
- Each lower level module (eg. 3GL): 10
- The sum of all of the above represents the NOP.

COCOMO 2: after prototyping

- Estimate the total lines of code (ESLOC)
- Takes into account
  - Requirements instability
  - Possibilities of code reuse

COCOMO 2: Influences on Costs

- Product attributes
  - Safety, module complexity, size of user manual, size of the required database, amount of reusable components
- Platform attributes
  - Constraints referring to execution time; platform volatility, memory constraints

COCOMO 2: Influences on Costs (cont.)

- Personnel attributes
  - Analyst experience; developer experience; personnel continuity; knowledge of the domain of the problem to be solved with regards to analysts and developers; knowledge of the programming language and development tools

- Project attributes
  - Required tools; distance between development teams (eg. different countries) and quality of communication; Development plan compression

The Planning Game

- A planning method from Extreme Programming (XP)
- A method used to manage the negotiation between the engineering team (Development) and the stakeholders (Business)
  - Treats the planning process as a game
  - playing pieces are “user stories” written on cards
  - the goal is to assign value to stories and put them into production over time

The Planning Game

- Unlike other planning methodologies, it does not require a documented description of the scope of the project to be estimated
- The Planning Game combines
  - estimation
  - identifying the scope of the project
  - Identifying the tasks required to complete the software

The Planning Game

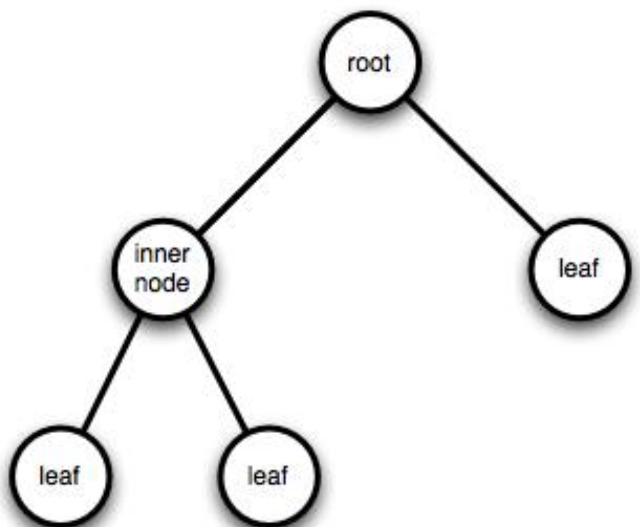
- The planning process is highly iterative. Each iterations looks like this:
  - **Scope** is established by having Development and Business work together to interactively write the stories.
  - Each **story** is given an estimate of 1, 2, or 3 weeks.
    - Larger stories are split into multiple iterations
  - Business is given an opportunity to steer the project between iterations.
  - The **estimates** are created by the programmers, based on the stories that are created.
  - **Commitments** are agreed upon

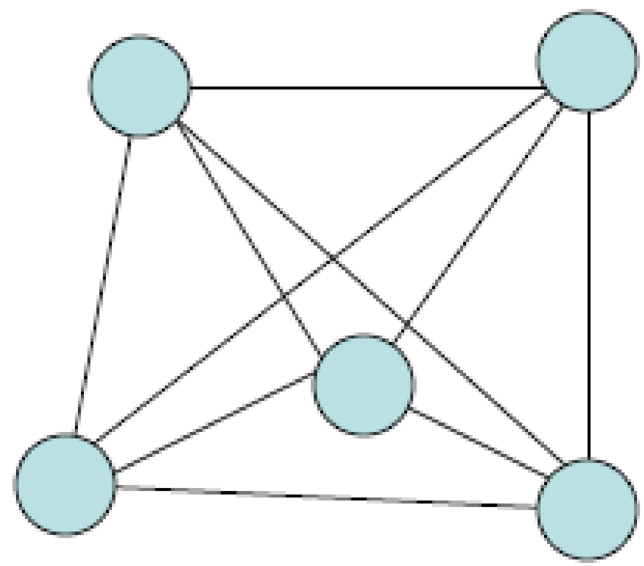
Distributing Workforce Over Time

- 20 PM. Are the following correct?
  - 20 people working 1 month
  - 4 people working 5 months
  - 1 person working 20 months
- Individual productivity decreases as team size increases
  - Communication overhead
  - On adding new members, productivity decreases initially
- *Adding people to a team behind of schedule makes that project more behind schedule.* (Brooks' law)

Distributing Workforce Over Time (2)

- For a team with P members, one can have between P-1 and P(P-1)/2 communication channels
- Each channel is a decrease in efficiency





How Not to Plan and Estimate Costs

- We have 12 months to finish the job, so it will take 12 months.
- A competitor asked for $1.000.000. We wil ask for $900.000.
- The client budget is $500.000. That will be the exact cost of development.
- Development takes 1 year, but we say it will take 10 months. A delay of 2 months is not important...

Problems with Metrics

- Lack of accuracy
- Employee pushback
- Use for other purposes than intended
- Animosity within the development team

Copyright

- The rights enjoyed by authors with regards to their work;
- Copyright is the instrument of protection of authors and their work;
- *Copyright gives the creator of an original work exclusive right for a certain time period in relation to that work, including its* ***publication, distribution and adaptation****; after which time the work is said to enter the public domain.*

Exclusive rights

- Several exclusive rights typically attach to the holder of a copyright:
  - to produce copies or reproductions of the work and to sell those copies (mechanical rights; including, sometimes, electronic copies: distribution rights)
  - to create derivative works (works that adapt the original work)
  - to perform or display the work publicly (performance rights)
  - to sell or assign these rights to others
  - to transmit or display by radio or video (broadcasting rights)

Bibliography

- Extreme Programming Explained, by Kent Beck (Addison Wesley, 2000)
- Applied Software Project Management, by Andrew Stellman and Jennifer Greene (O'Reilly, 2006)

Links

- **Bug Life Cycle:** http://www.buzzle.com/editorials/4-6-2005-68177.asp, http://qastation.wordpress.com/2008/06/13/process-for-bug-life-cycle/
- **COCOMO:** http://en.wikipedia.org/wiki/COCOMO
- Curs 12, Ovidiu si Adriana Gheorghies: http://www.info.uaic.ro/~ogh/files/ip/curs-12.pdf

Links

- Software Quality Assurance: http://satc.gsfc.nasa.gov/assure/agbsec3.txt
- Software Testing: http://en.wikipedia.org/wiki/Software_testing
- GUI Software Testing: http://en.wikipedia.org/wiki/GUI_software_testing
- Regression Testing: http://en.wikipedia.org/wiki/Regression_testing
- Junit Test Example: http://www.cs.unc.edu/~weiss/COMP401/s08-27-JUnitTestExample.doc
- https://www.test-institute.org/What_is_Software_Quality_Assurance.php

Coding Style – Links

- C++:
  - http://www.chris-lott.org/resources/cstyle/
  - http://geosoft.no/development/cppstyle.html
- Java:
  - http://java.sun.com/docs/codeconv/
  - http://geosoft.no/development/javastyle.html